Chapter 3

← Strings →

Strings is a data type in Python.
Strings is a sequence of characters unclosed in quotes.

We can primarily, write a string in these three ways.

1> Single quoted strings → a = 'Happ'
2> Double quoted strings → b = "Sad"
3> Triple quoted strings → c = """Sick"""

* String slicing

A string in Python can be sliced for getting a Part of the string. Consider the following string:

name = "

H	a	r	r	y
0	1	2	3	4
(-5)	(-4)	(-3)	(-2)	(-1)

" ⇒ length = 5

The index in a string starts from 0 to (length-1) in Python. In order to slice a string, we use the following syntax:

SL = name[index_start : index_end]

First index included last index is not included

SL[0:3] returns "Har" → character from 0 to 3
SL[1:3] returns "ar" → character from 1 to 3

Negative indices:

Negative indices can also be used as shown in the figure above -1 corresponds to the (length-1) index -2 to (length-2).

Slicing with skip value :-
We can provide a skip value as a part of our slice like like this:

Word = "amaging"
word[1:6:2] → 'mzn'

Other advanced slicing techniques

word = "amaging"
word[:7] → Word[0:7] → 'amaging'
word[0:] → word[0:7] → 'amaging'

String function → This function returns the length of the string

1> len()

len("Harry") → returns 5

2> name.endswith("rry") → This function tells whether the variable string ends with string "rry" or not. if string is "Harry", it returns True for "rry" since Harry ends with rry

3> String.count("c") → counts the total number of occurance of any character

4> string.capitalize() → This function capalilizes the first character of a given string.

5> string.find(word) → This function finds a word and returns the index of first occurence of that word in the string.

6> String.replace(old word, new word) - This function replace the old word with new word in the entered string.

Escape Sequence characters

Sequence of characters after backlash '\'

Escape Sequence character comprises of more than one characters but represents one character when (characters) used whithin the strings.

Example :- \n , \t , \' , \\ etc

newline Tab singlequate backlash

Chapter 3 - Practice set

1. Write a Program to display a user entered name followed by Good Afternoon using input() function.
2. Write a Program to fill in a letter template given below with name and date.

```
letter = """ Dear <|Name|>,
             You are Selected!
             <|Date|> """
```

3. Write a program to delet double spaces in a string.
4. Replace the double spaces from 3 with single spaces.
5. Write a program to format the following letter using escape sequence characters.

```
letter = "Dear Harry, This Python course
          is nice. Thanks!"
```

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Chapter-3_Strings

EXPLORER

CHAPTER-3_STRINGS

- .ipynb_checkpoints
- .venv
- .vscode
- main.py
- practiceset.ipynb
- str_func.py

main.py | str_func.py | practiceset.ipynb

main.py > ...

```python
name="virendra" # this is a string

# sl=name[0:4]
# print(sl) # string sorting
# character1=name[1]
# print(len(name))
ng=name[-8:]

arr="harry"
print(arr[-5:]) # negative indexing equivalent to positive indexing
print(arr[:5])
#print(ng)
name2="virendra kumar singh"
#skip slicing
print(len(name))
skip=name[0:8:2]
print(skip)

skip2=name2[::5]
print(skip2)
```

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Chapter-3_Strings

EXPLORER

CHAPTER-3_STRINGS

- .ipynb_checkpoints
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- 1.png
- main.py
- practiceset.ipynb
- str_func.py

main.py | str_func.py | practiceset.ipynb

practiceset.ipynb > #Q-5 write a program to format the following letter using escape

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```python
# Q-1 write a program to display user entered name followed by Good Afternoon using input() function
name = input("Enter your name: ")
print(f"good afternoon {name}")
```

[ ]

```python
# Q-2 write a program to fill in a letter template given below with name and date
letter = '''Dear <|NAME|>,
you are selected!
<|DATE|>'''
name = input("Enter your name: ")
date=input("enter date: ")
print(letter.replace("<|NAME|>",name).replace("<|DATE|>",date))
```

[3]

```
Dear virendra kashyap,
you are selected!
15 november 2000
```

```python
# Q-3 write a program to delete double spaces in a string
st1="my name  is  virendra  kashyap"
print(st1.replace("  ","_"))
```

[6]

OUTLINE

TIMELINE

0 0 Spaces: 4 LF Cell